Qistas: Pre-Registration of the First Iraqi Pilot

OSF Preregistration form (eight-page version in force in October 2026), text as deposited in draft on 24 October 2026, version 1.1

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This document reproduces, field by field and in the form's own order, the text entered in the OSF Preregistration draft created on 24 October 2026 (draft identifier 6adbd6adc7dff1ac850dadf0). Version 1 of this document (15 October 2026) followed the older 24-field layout of the same template; the registry now presents the template as eight pages with different field names, so version 1.1 re-keys the same commitments to the fields as they stand. Where the two versions differ in wording, the deposited text governs. The registration is submitted on the day the protocol paper is sent to arXiv, so that the registration link and date appear in the paper; until then the paper cites "OSF registration, DOI to appear". No model has been run at the time of registration.

Page 1: Metadata

Title

Qistas pilot: differential refusal, register effects, and stereotype reliance of large language models across Iraqi social constituencies

Description

Qistas is a protocol for evaluating how large language models (LLMs) treat the social, confessional, and regional constituencies of a society. It measures procedural properties only: whether a model's treatment of two prompts that differ in one group variable is symmetric, whether it refuses one arm of such a pair more than the other, and whether it relies on a stereotype when a question has no supportable answer. It never adjudicates contested truth and never ranks constituencies. The pilot registered here is the first instantiation, for Iraq, in Arabic, with Modern Standard Arabic and Iraqi Arabic as a register factor. The pilot evaluates ten to twelve systems on the automatically scored axes of the protocol (differential refusal, register twins, and stereotype reliance). The quality-gap axis (pairwise comparison of generated text by human raters) is run and archived but not scored under this registration; see "Other".

Contributors

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Affiliated institutions

None (the University of Misan is not an OSF institutional member).

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Subjects

Social and Behavioral Sciences: Linguistics. Physical Sciences and Mathematics: Computer Sciences: Artificial Intelligence and Robotics.

Tags

large language models; fairness evaluation; bias; Arabic; Iraqi Arabic; preregistration; differential refusal; stereotype reliance; counterfactual evaluation.

Page 2: Overview

Research questions or hypotheses

All hypotheses are tested per system; the pilot makes no claim about “models in general”. Each system is run twice, independently. Run 1 carries the registered test and run 2 is a pre-registered direct replication. A hypothesis is “supported for a system” only if the Holm-corrected test rejects the null in both runs; a rejection in one run only is reported as “not replicated”.

H1 (directional). On factorial templates (templates authored as semantically identical twins in Modern Standard Arabic and Iraqi Arabic), pooled across domains, the Iraqi-register twin is refused more often than its MSA twin. Null: the two discordant counts are equal in expectation. Statistic: exact McNemar test on template twins, one-sided in the direction “Iraqi refused more”. Family: one test per system.

H2 (two-sided). Differential refusal exists between the two arms of counterfactual pairs on at least one social axis. For each axis (regional origin, rural/urban origin, class, gender, tribal affiliation, marital status, ethno-regional origin, sect, religious community), pairs are pooled across domains. Null: discordant counts equal in expectation. Statistic: exact McNemar, two-sided. The “first arm” of every pair is the first-listed value in the authored value list, which is disclosed with the frozen bank; the direction of any detected asymmetry is reported. Family: systems x axes.

H3 (directional). Register interacts with axis: on factorial templates, the between-arm refusal difference is larger in the Iraqi register than in MSA. For each factorial template the statistic is the difference of two differences, (refusal of arm B minus arm A in Iraqi) minus (the same in MSA), with refusal coded 0/1 per item after aggregation over repetitions. Null: the mean of this difference is zero. Statistic: sign-flip permutation test clustered by template, one-sided in the direction “larger in Iraqi”. Family: one test per system.

H4 (directional). On ambiguous stereotype-reliance items, the stereotype-consistent share among committed answers exceeds 0.5. A "committed answer" is a choice of one of the two named persons rather than the "cannot be determined" option. Null: the share equals 0.5. Statistic: exact one-sided binomial test on committed answers. Validity gate, applied before testing: the system's accuracy on the disambiguated twins must be at least 0.80; otherwise its H4 result is reported as "uninterpretable: fails disambiguated accuracy gate" and no test is run. Family: one test per system.

Not tested confirmatorily: any single-domain comparison; any comparison between two named systems; any statement about the quality gap on the counterfactual-pair axis; any comparison across systems of the size of an effect. These are reported descriptively with intervals.

Foreknowledge of data or evidence

Data does not yet exist. No part of the data that will be used for this analysis plan exists, and no part will be generated until after this plan is registered.

Explanation of foreknowledge and managing unintended influences

No model output exists at the time of registration. Before the registered runs, the pipeline will be exercised on two systems with a handful of throw-away prompts solely to verify connectivity, logging, and the verification certificate; those outputs are not part of the item bank, are not scored, and are discarded. The analysis script is published before the runs and executed as published.

Page 3: Research Design

Study type

Non-randomized study. There is no random assignment of systems to conditions: every system receives every item, and the manipulation of the group variable is fixed within each pair by authoring.

Intention for causal interpretation

Indirect inference on causal relationship(s). The counterfactual pair isolates the effect of the swapped variable on the model's behaviour, but the pilot makes no claim about the mechanism (the emergence stage of a detected asymmetry is a hypothesis for later work, not a registered test).

Blinding of experimental treatments

Subjects will not be aware of the assigned treatment during data collection (the systems are not informed of the measurement and the prompts contain no reference to it). Researchers or observers who code or interpret data for analysis will not be aware of the assigned treatments during coding (human raters and coders are blind to the identity of the system).

Additional blinding during research or analysis

Prompts contain no reference to the measurement (authoring rule R-005: no prompt may name the criterion being measured). Human raters and coders are blind to the identity of the system whose output they adjudicate; outputs are presented under random identifiers. The authors are not blind to system identity during analysis; the analysis is fully scripted and the script is published before the runs and executed as published.

Study design

Within-system paired design. Every item belongs to a template group; a counterfactual pair consists of two items identical except for one group variable. Factorial templates additionally carry an MSA/Iraqi register twin, giving a 2 (group arm) x 2 (register) within-template factorial. Stereotype-reliance items follow the BBQ ambiguous/disambiguated-twin design. Every system receives every item under identical conditions (identical prompts, identical decoding parameters declared in the run certificate, identical retry ladder). All comparisons are within-system and paired at the item or template level; there are no between-subject arms.

Randomization

Order of items within a run is randomised with a seeded generator so that repetitions of the same item are not adjacent. Answer-option positions in stereotype-reliance items are balanced by design at authoring and audited against uniformity with a chi-squared test before analysis; options are not shuffled at run time. The random seed is recorded in each run certificate.

Page 4: Sampling

Data collection procedures

The item bank is authored natively in Arabic under the published authoring guides, passes the automated structural audit (zero hard violations) and a checklist-driven human semantic review, and is then frozen; the frozen file's SHA-256 digest and the final system list are posted as a dated addendum to this registration (as an uploaded file or a registration update) before any run begins. Each system is queried through its provider API or through an OpenAI-compatible self-hosted endpoint, three times per item, in two complete independent runs, with the raw response envelope archived for every attempt. Refusal is detected automatically from the response text and, where available, from the provider's structured refusal fields, with human verification on a 25% sample of refusal-axis responses. Stereotype-reliance items are scored by exact answer-letter match.

Sample size

Units: systems (ten to twelve; the final list is fixed in the addendum) and, within each system, items. Item bank at freeze: 100 counterfactual pairs (200 items); 60 ambiguous stereotype-reliance items with their 60 disambiguated twins; 50 differential-refusal pairs (100 items); 40 to 50 terminological-framing items; between 40% and 50% of templates carry a register twin.

Every system receives the same items, so sample sizes are identical across systems. Three repetitions per item and two runs per system.

Sample size rationale

Computed minimum detectable effects at alpha = 0.05 and power 0.80, with a design effect of 1.3 applied to pair-level statistics to account for two pairs per factorial template at an assumed intra-template correlation of 0.3: counterfactual pairs, paired d_z of about 0.32 at the axis level (about 0.52 for a 40-pair single-domain slice, which is why single-domain slices are descriptive); differential refusal, with exact McNemar power a function of discordant pairs, a split of about 84:16 among 15 discordant pairs (30% discordance at 50 pairs), with the full dependence tabulated in the protocol paper; stereotype reliance, a committed share of 0.72 at a 50% commitment rate (30 trials), 0.66 at full commitment. The refusal axis was enlarged from 30 to 50 pairs because at 30 pairs it could detect only near-total one-sidedness. Any "no evidence" result is reported together with these bounds.

Starting and stopping rules

Start: data collection begins only after the item bank is frozen and its SHA-256 digest and the system list are posted as the dated addendum; a pipeline check on two systems with throw-away prompts precedes it and produces no analysed data. Stop: data collection stops when both runs of every listed system are complete, or when a system has exhausted the retry ladder on more than 5% of its items, in which case that system is reported as "incomplete" with its completed items analysed and the incompleteness disclosed. No interim analysis is performed and no system is added or removed after the addendum.

Page 5: Variables

Manipulated variables

Four within-system manipulations, all assigned by design (no randomization of assignment):

1. Group arm of the counterfactual pair: the swapped value on the declared axis (regional origin, rural/urban origin, class, gender, tribal affiliation, marital status, parenthood, ethno-regional origin, sect, religious community). Two levels per pair; the "first arm" is the first-listed value in the authored value list, disclosed with the frozen bank.
2. Register, on factorial templates only: Modern Standard Arabic versus Iraqi Arabic, as semantically identical twins.
3. Context, on stereotype-reliance items: ambiguous (no supportable answer) versus disambiguated (a single decisive sentence replaces the no-information sentence).
4. Disambiguation direction, on stereotype-reliance templates: the decisive fact resolves with or against the prevalent stereotype, balanced across the bank.

Measured variables

Per item and repetition: refusal (explicit or veiled, under the coding-guide definitions; coded 0/1); for stereotype-reliance items, the option chosen and whether it is the stereotype-

consistent person, the other person, or “cannot be determined”; for disambiguated twins, correctness of the chosen option; technical outcome of the attempt (completed, empty, truncated, failed after the retry ladder). Variables used for subsetting: system identity, run number (1 or 2), test type, domain, axis, register, sensitivity band of refusal pairs, template identifier.

Indices

Per item: refusal is the majority outcome over the three repetitions (refused in at least two of three); the stereotype-reliance choice is the modal choice over repetitions, with ties recorded as “cannot be determined”. Per system and axis: refusal rate per arm with a 95% Wilson interval and a template-clustered bootstrap interval; discordant-pair counts for the McNemar tests (pairs where exactly one arm is refused); for factorial templates, the difference of differences (refusal of arm B minus arm A in Iraqi) minus (the same in MSA); commitment rate on ambiguous items (share of answers naming one of the two persons); stereotype-consistent share among committed answers; disambiguated accuracy (share of disambiguated twins answered with the keyed option). Between runs: per-axis agreement of item-level outcomes.

Page 6: Analysis Plan

Statistical models

H1 and H2: exact McNemar test on discordant pairs, that is an exact binomial test of the discordant split against 0.5; one-sided for H1 (direction: Iraqi twin refused more), two-sided for H2. H3: sign-flip permutation test on the per-template difference of differences, 10,000 resamples, template as the unit of resampling, one-sided (direction: larger between-arm difference in Iraqi). H4: exact binomial test of the committed stereotype-consistent count against 0.5, one-sided, run only after the validity gate (disambiguated accuracy at least 0.80); a system failing the gate is reported as “uninterpretable” and no H4 test is run for it. Every proportion is reported with a 95% Wilson interval; every proportion computed over template-generated items is additionally reported with a template-clustered bootstrap interval (2,000 resamples). Effect sizes: the discordant-pair odds for McNemar tests, paired d_z for the permutation test, and the share itself with its interval for H4. No omnibus tests, no regression models, no covariates. The analysis code is the published metrics library and protocol analyser at the repository commit named in the addendum, and is executed as published.

Transformations

Refusal is coded 0/1 per attempt under the coding-guide definitions (explicit and veiled refusal both coded 1). Stereotype-reliance answers are coded as one of three categories (stereotype-consistent person, other person, cannot be determined) by exact answer-letter match against the keyed option order. Aggregation over the three repetitions follows the rules stated under Indices (majority for refusal, mode for choice, ties to “cannot be determined”). No other transformation, centering or recoding is performed.

Inference criteria

alpha = 0.05 after Holm correction within each family. Families: H1, one test per system (family size = number of systems); H2, systems x axes; H3, one test per system; H4, one test per system. Sidedness: H1, H3 and H4 one-sided in the stated direction; H2 two-sided. Sidedness will not be changed after data are seen. A hypothesis is "supported for a system" only if the Holm-corrected test rejects the null in both independent runs; rejection in one run only is reported as "not replicated". No claim is made about models in general, no system is compared with another confirmatorily, and no composite score across domains or axes is computed.

Data inclusion and exclusion

All items of the frozen bank and all listed systems are included. Attempts that end empty or truncated by the provider's envelope after the retry ladder are flagged and excluded from stereotype-reliance scoring; in refusal metrics they count as non-refusals only if the response contains substantive content before truncation, otherwise as technical failures excluded from both. Deterministic provider-side refusals are counted as refusals, not technical failures, and are not re-run. Items withdrawn after the freeze (for example on discovery of an authoring defect) are withdrawn publicly with their identifier, and their effect on every affected statistic is reported. There is no outlier handling: the outcomes are binary or categorical.

Missing data

An item with no completed attempt in a run is missing for that run, and its pair or template is dropped from the paired statistics of that run; the count of dropped units is reported per axis and per system. A system that exhausts the retry ladder on more than 5% of its items is reported as "incomplete" with its completed items analysed and the incompleteness disclosed. No imputation.

Other planned analysis

Exploratory, reported with intervals and not corrected: single-domain slices; descriptive comparison of refusal rates across the three sensitivity bands of refusal pairs; distributions of watchlist terms on terminological-framing items coded by two human coders, with Krippendorff's alpha; explicit versus veiled refusal; between-run disagreement patterns; descriptive quality-gap comparisons on the counterfactual-pair axis if independent raters become available. All are labelled exploratory in the report.

Page 7: Other

Context and additional information

Protocol and documents. This registration covers the first instantiation of Qistas, a portable protocol for fairness evaluation of large language models, for Iraq. The protocol paper (Qistas: A Pre-Registered, Portable Protocol for Fairness Evaluation of Large Language Models, Alseelawi and Aljumaily, 2026, to appear on arXiv) describes the taxonomy, the four test types, the authoring guides, the rulings log, the structural audit and the power analysis; the methodology,

taxonomy and authoring-guide documents and the reference implementation (metrics library, runner, analyser, audit, review-notebook builder, rulings log, published tier of the item bank) will be published in a public repository whose URL and commit hash are named in the dated addendum. Qistas inherits its runner, integrity protocol (dated snapshots, SHA-256 verification certificates) and dual-review practice from the sibling project Mizan (arXiv:2609.13980).

Item bank and addendum. The bank is not frozen at registration. Its sizes are registered as targets, and the power statements are given as functions of size; the frozen bank's SHA-256 digest, the exact per-type counts, the first-arm order of every pair, the final system list with versions and decoding parameters, and the repository commit of the analysis code are posted as a dated addendum to this registration before any registered run begins. Half of the templates are held in a sealed tier that is never published; the published half suffices to verify the structure of the bank.

Second-author audit. Items enter the bank on the first author's checklist verdict; before the freeze the second author audits a stratified sample of 20 to 25 percent of template groups without access to the first author's verdicts. One documented design disagreement between the authors (whether naming the two groups in an ambiguous-choice item is itself a cue, and whether naming a sect in a refusal pair changes its legitimacy) is recorded in the public rulings log and carried to that audit; it does not change the registered design.

Axes run but not scored under this registration. The quality-gap axis (pairwise comparison of generated text on counterfactual pairs by human raters against a rubric) is run and archived but is not scored here; the rubric, its calibration sample and the agreement threshold will be registered as a dated amendment before any quality adjudication begins, and only after independent raters are secured. The terminological-framing items are coded by the two authors (author-coders, no independent raters) and reported descriptively with coder agreement; no confirmatory hypothesis is registered for them. Human adjudication in the pilot covers all framing items, a 25% sample of refusal-axis responses and, after the amendment, a 15% calibration sample of counterfactual-pair outputs, each by two raters.

Reporting constraints. Results are reported per system and per domain as scorecards with intervals; no ranking of systems, no composite score across domains or axes, and no characterisation of any system as "neutral" or "biased" in the absolute. Systems whose intervals overlap are reported as a single statistical tie tier, and every symmetry metric is displayed with its companion response-rate indicator. Fairness is understood procedurally (symmetry of treatment, balance, transparency), never as a verdict on contested truth, and the protocol never ranks the constituencies it studies.

Deviations. Any departure from this registration is recorded in a public deviation appendix with date, reason and effect, and in the project's rulings log.

Competing interests. Both authors are members of the national technical team building Iraq's first large language model. Any system in which an author has a role is marked as such on every scorecard.

Disclosure. An LLM research assistant was used throughout the project for drafting and tooling, including the drafting of this registration from the authors' decisions; every design decision

and every item verdict is the authors' own and is recorded in the rulings log. No model has been evaluated at the time of registration.

Addendum (to be posted before the first registered run)

Date of freeze; SHA-256 digest of the frozen internal bank and of the published tier; per-type item counts; first-arm order of every pair; system list with provider, version string, decoding parameters and endpoint type; repository URL and commit hash of the analysis code; random seeds of both runs.